

ORIGINAL ARTICLE

Food Microbiology and Safety

Polygonatum cyrtonema Polysaccharide-Stabilized High Internal Phase Emulsions Enhance the Gastrointestinal Survival and Thermal Stability of *Lactobacillus acidophilus*

Zeao Wang¹  | Zhifa Zhang² | Chunyan Peng³ | Yihui Dai⁴  | Yi Huang¹ | Xiaoqiang Chen^{1,4} 

¹ College of Life Science and Health Engineering, Hubei University of Technology, Wuhan, China | ² Department of Anesthesiology, Tongji Hospital, Tongji Medical College, Huazhong University of Science and Technology, Wuhan, China | ³ Department of Internal Medicine, Maternal and Child Health Hospital of Hubei Province, Wuhan, China | ⁴ College of Health Science and Engineering, Hubei University, Wuhan, China

Correspondence: Xiaoqiang Chen (biomed528@163.com)

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ABSTRACT

This study successfully developed a high internal phase emulsion (HIPE) stabilized by *Polygonatum cyrtonema* polysaccharide (PCP) for the efficient delivery of *Lactobacillus acidophilus* (*La*). Characterization of the extracted polysaccharide revealed an average molecular weight of approximately 1.27×10^5 Da. Ultraviolet–visible (UV–Vis) scanning and Fourier-transformed infrared (FTIR) spectra confirmed the high purity of PCP and revealed its characteristic pyranose and furanose ring structures. Atomic force microscopy observation showed a non-helical, entangled network morphology. The emulsion system maintained a probiotic survival rate of 63.75% under simulated gastrointestinal digestion, retained 50.2% viability after heat treatment at 70°C, and preserved stable bacterial counts after 28 days of storage at 4°C, demonstrating significantly superior protection ($p < 0.05$) compared to the gum arabic-based system. Mechanistic analysis indicated that the amphiphilic structure of PCP enables the formation of dense interfacial films and gel networks, effectively shielding the probiotics from environmental stress. These findings confirm the potential of PCP-based HIPEs as a green and efficient probiotic delivery carrier for functional food applications.

1 | Introduction

As a distinguished homologous medicinal and edible plant, *Polygonatum cyrtonema* contains a wealth of bioactive secondary metabolites, notably polyphenols and polysaccharides, with the latter serving as the primary driver of its therapeutic potential. These macromolecules have garnered significant scientific interest due to their versatile physiological benefits, particularly their roles in anti-oxidation, anti-inflammation, and the regulation of metabolic homeostasis (Li et al. 2025). *P. cyrtonema* polysaccha-

ride (PCP) possesses a backbone rich in hydrophilic hydroxyl groups, while spectroscopic analyses suggest the presence of hydrophobic microdomains, likely arising from specific glycosidic linkages or intramolecular arrangements (Wang et al. 2022). This inherent amphiphilicity, combined with a moderately high molecular weight (approximately 10^5 Da), provides a molecular foundation for its potential to adsorb at oil–water interfaces and form viscoelastic films, thereby supporting its function as a natural emulsifier for constructing colloidal systems (Zhang et al. 2025).

Zeao Wang and Zhifa Zhang contributed equally to this work.

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High internal phase emulsions (HIPEs), characterized by an internal phase volume exceeding 74%, offer distinct advantages over conventional emulsions. Their densely packed microstructure and the resulting three-dimensional network impart remarkable physical stability against coalescence, flocculation, and Ostwald ripening (Yang et al. 2015). In recent years, polysaccharide-based stabilizers have been increasingly explored for HIPE formulation due to their biocompatibility, sustainability, and often unique interfacial behavior. Building on this trend, the structural attributes of PCP position it as a viable candidate for stabilizing such systems. Furthermore, the protective microenvironment provided by HIPEs makes them particularly suitable for encapsulating sensitive bioactive agents (Jie 2023). This is especially relevant for probiotics, which often suffer from low viability during processing, storage, and gastrointestinal transit. Encapsulation within a polysaccharide-stabilized HIPE matrix can physically shield probiotic cells from environmental stresses, thereby enhancing their survival and targeted delivery—a strategy already demonstrated with other polysaccharides, supporting its feasibility for PCP-based systems (Garti and Leser 2001).

Despite the promising bioactive properties of PCP, its potential as a specialized emulsifier for stabilizing probiotic-loaded HIPEs remains largely unexplored. Given that current commercial delivery systems often fail to maintain sufficient microbial viability under harsh physiological conditions (Jiang et al. 2020), there is a critical need for robust, plant-based encapsulation matrices. Therefore, the present study aimed to develop PCP-stabilized HIPEs for the effective encapsulation and protection of *Lactobacillus acidophilus* (*La*) (Yin et al. 2025). Guided by the unique non-helical, entangled network morphology of PCP, we first systematically investigated the influence of PCP concentration on HIPE formation to identify the optimal interfacial formulation. HIPEs were then prepared using medium-chain triglyceride as the oil phase and an aqueous suspension containing both *La* and PCP. The resulting emulsions were characterized in terms of macroscopic appearance, microstructure, droplet size, and size distribution. Their stability and protective efficacy were systematically evaluated through storage tests and in vitro simulated gastrointestinal digestion. Through elucidating the structure–function relationship of PCP in HIPE stabilization and rigorously validating its protective efficacy against gastrointestinal, thermal, and storage stressors, this research positions PCP as a promising, eco-friendly biomaterial for the high-performance delivery of probiotics within the functional food sector.

2 | Materials and Methods

2.1 | Materials

PC was purchased from Mingtong Pharmaceutical Co. Ltd. (Bozhou, China). MCT (primarily consisting of caprylic (C8) and capric (C10) triglycerides) was obtained from Guangzhou Yongsheng Industrial and Trading Co. Ltd. *La* was procured from the China Center for Type Culture Collection (Wuhan). Standard substances and other analytical reagents were purchased from Sigma-Aldrich (St. Louis, MO, USA). Ultrapure water was used throughout the experiment for solution and emulsion preparation.

2.2 | Preparation of PCP

Raw *P. cyrtonema* was pulverized and extracted with deionized water (1:20 w/v) at 85°C for 2 h (Liu et al. 2010). The extract was filtered and centrifuged (CR21N, Hitachi, Osaka, Japan) at 7155 × g for 10 min, and the supernatant was concentrated via membrane separation (MWCO: 3500 Da) at room temperature. Polysaccharides were precipitated by adding four volumes of ethanol for 6 h, then recovered by centrifugation (CR21N, Hitachi, Osaka, Japan). Lyophilization of the sample was performed using a freeze dryer (FreeZone, Labconco, Kansas City, MO, USA) to yield the final PCP product.

2.3 | Characterization of PCP

2.3.1 | Molecular Weight Distribution

Gel permeation chromatography coupled with multi-angle laser light scattering (GPC-MALLS) was employed to determine the molecular weight and distribution of PCP (DAWN HELEOS II, Wyatt Technology Corp., Santa Barbara, CA, USA) (Bai et al. 2021). A 1.0 mg/mL PCP solution was prepared using a 0.20 mol/L NaCl solution, centrifuged (4°C, 9056 × g, 20 min), and filtered through a 0.22 µm membrane for analysis. The HPLC system, equipped with a gel chromatography column, was maintained at 25°C ± 0.2°C, with the mobile phase delivered at a flow rate of 0.40 mL/min and an injection volume of 200 µL. During chromatographic separation, the absorbance of the eluent was measured using a MALLS-RI-LS detector. The weight-average molecular weight (M_w), number-average molecular weight (M_n), and polydispersity index (PDI) were obtained.

2.3.2 | Ultraviolet-Visible Spectroscopy (UV-Vis) and Fourier-Transformed Infrared (FTIR) Spectroscopy

An appropriate amount of PCP was dissolved in ultrapure water to yield a 100 µg/mL solution, followed by centrifugation at 9056 × g for 6 min before further use. UV-Vis spectra were recorded in the range of 200–800 nm using a spectrophotometer (T6, Persee General Instrument Co. Ltd., Beijing, China), with a bovine serum albumin (BSA) solution as a control.

For FTIR analysis (Nicolet iS10, Thermo Fisher Scientific, Waltham, MA, USA), PCP was thoroughly mixed with potassium bromide (KBr) powder at a ratio of 1:100 (w/w), ground into a fine powder, and pressed into pellets. Infrared spectra were acquired over a frequency range of 4000–400 cm⁻¹ (Zeng et al. 2024).

2.3.3 | Atomic Force Microscopy (AFM)

A 1 mg/mL PCP solution was centrifuged (9056 × g, 6 min) and stirred for 12 h. The solution was then diluted to 10 µg/mL and stirred for another 12 h. A small amount of the solution was dropped onto freshly cleaved mica and air-dried at room temperature for AFM (Dimension, Bruker Rheinstetten, Germany) observation (Zhang et al. 2023).

2.4 | Interfacial Properties

The interfacial tension of PCP aqueous solutions at different concentrations at the oil–water interface was measured using a Tracker interfacial rheometer (Tracker, Teclis Scientific, Lyon, France). PCP aqueous solutions at concentrations of 0.3 and 0.5 mg/mL were prepared and centrifuged at $9056 \times g$ for 15 min. Subsequently, an aliquot of approximately 10 mL was transferred into a cuvette and positioned within the instrument's sample chamber. A U-shaped needle containing MCT was immersed into the sample solution. An oil droplet of approximately 10 μ L was dispensed by precisely controlling the syringe. The oscillatory measurement was then monitored over a duration of 12,000 s, maintained at an amplitude of 10% and a frequency of 0.10 Hz (Yun et al. 2022).

2.5 | Preparation of HIPEs With Different PCP Concentrations

To prepare the HIPEs, varying amounts of PCP (0.10, 0.15, 0.20, and 0.25 g) were first dispersed in 2.0 g of ultrapure water using a high-speed shear homogenizer (FA25, FLUKO, Shanghai, China) coupled with vortex agitation. Each resulting aqueous phase was then blended with 8.0 g of MCT to yield mixtures with PCP (as emulsifier) concentrations of 1.0, 1.5, 2.0, and 2.5 wt% (w/w), respectively, maintaining a constant oil phase (MCT) content of 80 wt%. The HIPEs were subsequently fabricated by homogenizing these mixtures at 16,000 rpm for 40 s. The final emulsions were stored at 25°C for further characterization.

2.5.1 | Morphology of HIPEs

Freshly prepared HIPEs were stored at 25°C for 24 h to allow system equilibration. The macroscopic appearance of the HIPEs was captured via digital photography. Meanwhile, the microstructure was immediately visualized after preparation using a polarized light microscope (Eclipse Ci-L, Nikon, Tokyo, Japan) to minimize potential structural changes over time. For microscopic observation, 3.0 μ L of each emulsion was placed on a slide, carefully covered, and examined (Chen et al. 2026).

2.5.2 | Average Particle Size and Distribution of HIPEs

The surface-weighted mean diameter ($D_{3,2}$) and particle size distribution of the emulsions were measured using a laser particle size analyzer (Mastersizer 2000, Malvern, England) with deionized water as the dispersant. The refractive indices of the continuous and dispersed phases were set to 1.330 and 1.441, respectively. The sample concentration was maintained at approximately 2.0%, and measurements were performed in triplicate (Yu et al. 2024).

2.6 | Cultivation of *La*

La was inoculated into de Man, Rogosa, and Sharpe (MRS) broth (Oxoid, Basingstoke, UK) and incubated at 37°C with shaking at 180 rpm for 24 h to prepare the inoculum. The inoculum was then

transferred to fresh MRS broth at a 1.00% (v/v) inoculation ratio and incubated under the same conditions for another 24 h.

2.6.1 | Encapsulation of *La* in HIPEs

A 2.0 wt% PCP solution was prepared with slight modifications (Su et al. 2018), specifically by omitting the thermal gelation step required for protein microgels, as PCP directly hydrates to form a viscous solution. The 24 h-cultured bacterial suspension was centrifuged ($4025 \times g$, 5 min), and the supernatant was discarded. The pellet was washed twice with PBS (0.01 mol, pH 7.4) and resuspended in the PCP solution. A 2.0 g aliquot of the bacterial suspension was mixed with 8.0 g of MCT to achieve a total system weight of 10.0 g. The mixture was homogenized using a high-speed shear mixer at 16,000 rpm for 40 s and stored at 25°C for further use. The encapsulation efficiency (EE) of *La* was considered to be nearly 100% due to the nature of the HIPE system. Given the immediate formation of a self-supporting gel structure and the high viscosity of the PCP-stabilized interface, the microorganisms were effectively entrapped within the network. To maintain the structural integrity of the HIPEs, no additional centrifugal separation of unencapsulated bacteria was performed. To enumerate the viable *La*, 1.0 g of the HIPEs was dispersed in 9.0 mL of sterile saline solution (0.85% NaCl) containing 0.1% (v/v) Tween 80. The mixture was vortexed thoroughly to disrupt the emulsion matrix, ensuring the complete release of the encapsulated cells prior to serial dilution and plate counting.

A 2.0 wt% gum arabic (GA) solution served as a control, because GA is a widely recognized gold-standard, highly branched natural polysaccharide emulsifier commonly utilized in the food industry for encapsulating sensitive bioactives.

2.6.2 | Morphology of HIPEs Containing *La*

Fluorescence microscopy (BZ-X800E, Keyence, Shanghai, China) was used to observe the internal structure of PCP-*La* HIPEs and GA-*La* HIPEs and confirm the encapsulation of *La* in the internal phase. Prior to HIPE preparation, the bacterial cells in the internal phase were stained with Rhodamine 123 and Nile red. For fluorescence labeling, 3.00 μ L aliquots of each fluorochrome were added to 2 mL of the bacterial suspension. The stained samples were then subjected to fluorescence imaging using green and red channels, respectively (Page et al. 2020).

2.7 | Simulated Gastrointestinal Digestion of Encapsulated *La*

Simulated gastrointestinal digestion was performed with modifications (Shi et al. 2021). Specifically, the ratio of the emulsion to the simulated digestive fluids was adjusted, and the concentration of bile salts was optimized. This modification was necessary because the HIPE system contains a remarkably high oil fraction (80%) and possesses a highly viscous gel network, requiring more digestive components to ensure adequate lipid digestion and structural disruption. Simulated gastric fluid (SGF) was prepared by adding 0.32 g of pepsin to 100 mL of sterile saline (0.85%

w/v, pH 2.0), resulting in an enzyme activity of 222.4 U/mL. Simulated intestinal fluid (SIF) was prepared by adding 0.3 g of bile salts to 100 mL of sterile saline (0.85% w/v, pH 7.4), followed by sterilization at 115°C for 30 min. To 100 mL of sterile bile salt solution, 10 mg of trypsin and 2.4 g of lipase were added to achieve enzyme activities of 250 and 720 U/mL, respectively.

For the digestion experiment, 1.0 mL of bacterial suspension or 1.0 g of emulsion was mixed with SGF at a 1:1 (v/v) ratio and incubated in a shaking water bath (ZHWY2102C, Shanghai, China) at 37°C and 50 rpm for 2 h. The initial microbial load of the 1.0 mL bacterial suspension was approximately 9.25 log CFU/mL, whereas that of the 1.0 g PCP-*La* HIPEs was approximately 8.71 log CFU/g. The lower initial count in the HIPE system is attributed to the dilution effect of the 80 wt% oil phase during the emulsification process. Subsequently, 1 mL of the mixture was combined with SIF at a 1:1 (v/v) ratio and incubated under the same conditions for another 2 h. After digestion, the HIPEs were centrifuged (11,180 × g, 10 min) to disrupt their structure and release the bacterial cells. The viability of the cells was determined using the plate counting method. Serial dilutions of the samples were plated on MRS agar and incubated at 37°C for 48 h before colony counting. Each sample was analyzed in triplicate. The survival rate (SR) was calculated using the following equation:

$$SR = \left(\frac{N}{N_0} \right) \times 100 \quad (1)$$

where N is the number of viable cells released from the emulsion after simulated digestion (log CFU/mL), and N_0 is the number of free cells before digestion (log CFU/mL).

2.8 | Heat Treatment of Encapsulated *La*

Free 1.0 mL of bacterial suspension and 1.0 g HIPEs were placed in sterile tubes and subjected to heat treatment in a water bath at 50°C, 60°C, and 70°C for 30 min. The HIPEs were then centrifuged (11,180 × g, 10 min) to release the bacterial cells, and viability was determined using the plate counting method as described above (Yiasmin et al. 2024).

2.9 | Storage Stability of Encapsulated *La*

2.9.1 | Viability of *La* During Storage

Free bacterial suspensions and *La*-loaded HIPEs were stored in sterile containers at 4°C for a 28-days stability study. The viability of *La* was monitored at predetermined intervals (1, 7, 14, 21, and 28 days), with the free cell suspension serving as a control group. This storage duration and temperature were selected to simulate the typical shelf life and distribution cycle of chilled probiotic-fortified functional foods (e.g., yogurt-like emulsions or spreads). All measurements were performed in triplicate to ensure reproducibility.

2.9.2 | Particle Size of HIPEs During Storage

The surface-weighted mean diameter ($D_{3,2}$) and particle size distribution of the emulsions were measured using a Mastersizer 2000 analyzer at 1, 7, 14, 21, and 28 days of storage. Each sample was analyzed in triplicate.

2.10 | Data Analysis

Data were analyzed and visualized using Origin 2019b, Microsoft Excel, and SPSS (version 25.0, IBM Corp., Armonk, NY, USA). All experiments were performed in triplicate ($n = 3$), and results were expressed as mean ± standard deviation (SD). Significant differences between groups were determined using one-way analysis of variance (ANOVA) followed by Tukey's post hoc test. A value of $p < 0.05$ was considered statistically significant.

3 | Results and Discussion

3.1 | Characterization of PCP

3.1.1 | Molecular Weight Distribution

Molecular weight is a critical structural parameter for polysaccharides, as it significantly influences their rheological properties, biological activities, and interfacial stability. M_w , M_n , and PDI were obtained. The PDI ratio reflects the breadth of the molecular weight distribution, with higher values indicating a broader distribution (Zou et al. 2024). As shown in the GPC chromatogram (Figure 1), PCP had an average molecular weight of 1.27×10^5 Da, with a PDI ratio of 5.531. The observation is consistent with that reported by Yi et al. (2019) for polysaccharides extracted from *Tetragium hemsleyanum*, where a similar polydispersity (PDI = 5.1) was observed. This high PDI value (5.531) is primarily attributed to the intrinsic molecular heterogeneity of complex natural glycans, resulting in inconsistent chain lengths and a broad molecular weight distribution.

3.1.2 | UV-Vis and FTIR Spectroscopy

The presence of free proteins in PCP was determined by comparing their UV characteristic absorption peaks at identical wavelengths. The spectrum of PCP (Figure 2A) exhibited no detectable absorption at 282 nm, which is the characteristic peak region for BSA, thereby confirming the absence of residual free proteins within the polysaccharide matrix. A discernible shoulder peak manifested at 257 nm, which may be attributed to the electronic transitions of inherent quinone-like pigments within the PCP (Chen et al. 2017).

FTIR was employed to characterize the functional groups of PCP. Previous studies have identified characteristic polysaccharide absorption bands (Figure 2B). The FTIR spectrum of PCP exhibited characteristic functional groups of polysaccharides. Specifically, the broad and intense absorption band at 3330.12 cm^{-1} was assigned to the O—H stretching vibrations of the pyranose rings (Li et al. 2020). The weaker band at 2937.66 cm^{-1} originated from the C—H stretching vibrations of the —CH₃

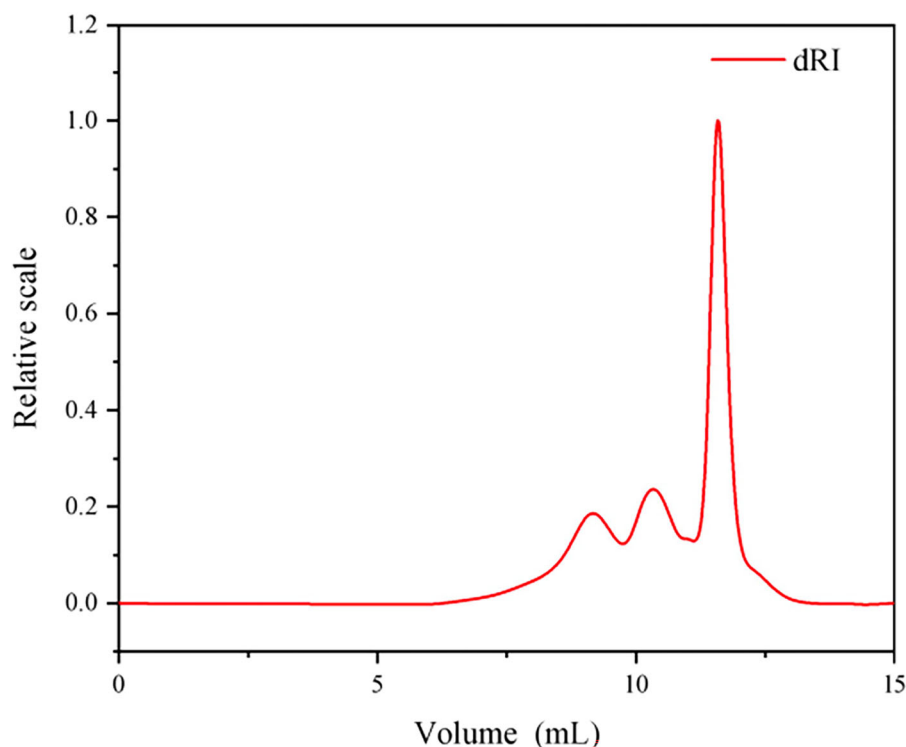


FIGURE 1 | The GPC chromatogram of PCP.

and —CH_2 groups (Cheng et al. 2024). The absorption peaks at 1729.91 and 1633.96 cm^{-1} were attributed to the C=O stretching vibrations, typically associated with esterified carbonyl groups and the asymmetric stretching of carboxylate groups (—COO—), respectively, which may also overlap with the bending vibrations of bound water. The peak at 1373.13 cm^{-1} corresponded to the C—H bending vibrations. Furthermore, two distinct peaks at 1244.40 and 1027.43 cm^{-1} were indicative of C—O—C glycosidic bond stretching within the pyranose rings. Finally, the absorption bands near 926.0 and 818.5 cm^{-1} confirmed the presence of furanose ring structures (Huang et al. 2024). Notably, the absence of typical protein absorption peaks at 1650 cm^{-1} (amide I) and 1550 cm^{-1} (amide II) confirms the low protein content in PCP. On the basis of the spectroscopic analysis, the results collectively indicate that PCP is primarily composed of polysaccharides. These polysaccharides exhibit characteristic pyranose and furanose ring structures and contain abundant hydroxyl and alkyl groups. No characteristic signals for free or conjugated proteins were detected in the spectra, confirming the high polysaccharide purity of PCP.

3.1.3 | Atomic Force Microscopy

AFM provides nanoscale topographical images of biopolymers by modulating probe-sample interaction forces (Wang and Nie 2019). As shown in Figure 3A, the 2D topography revealed PCP molecules exhibiting irregular spherical or coiled conformations with minor spherical aggregates, where polysaccharide chains were interwoven on the mica substrate. The 3D reconstruction (Figure 3B) demonstrated needle-like morphologies with measured heights exceeding those of individual polysaccharide chains, indicating branched and interconnected PCP structures.

This molecular entanglement likely results from strong intermolecular and intramolecular interactions mediated by hydroxyl and carboxyl groups in the acidic polysaccharides (Liu et al. 2016). Importantly, the measured heights (significantly below 15–50 nm) provide conclusive evidence that PCP lacks a triple-helical conformation (Chen, Sun, et al. 2023).

3.2 | Interfacial Properties

Interfacial rheology testing was employed to evaluate the oil-water interfacial properties of PCP, which are critical for assessing its emulsifying capacity. Upon forming the emulsion, PCP molecules rapidly accumulate at the oil-water interface, effectively lowering the interfacial energy and facilitating the production of smaller droplets (Chen, Niu, et al. 2023; Zhang et al. 2021). As shown in Figure 4, the interfacial tension of PCP decreases sharply at the initial stage and then continues to decline at an almost constant rate over time. Specifically, the oil-water interfacial tension of 0.3 mg/mL PCP decreases from 32.41 to 18.67 mN/m, whereas that of 0.5 mg/mL PCP significantly decreases from 31.27 to 17.53 mN/m ($p < 0.05$). As the concentration increases, the interfacial tension gradually decreases ($p < 0.05$), indicating faster diffusion of PCP at the oil-water interface. Higher concentrations of PCP possess more amphiphilic groups, thereby demonstrating stronger interfacial stability and adsorption capacity. This robust surface activity is fundamentally governed by the specific molecular features of PCP. The amphiphilicity required to rapidly anchor at the oil-water interface arises from the coexistence of abundant hydrophilic hydroxyl groups (3330.12 cm^{-1}) and hydrophobic microdomains formed by alkyl groups (2937.66 cm^{-1}) and esterified carbonyls (1729.91 cm^{-1}), as confirmed by the FTIR spectra.

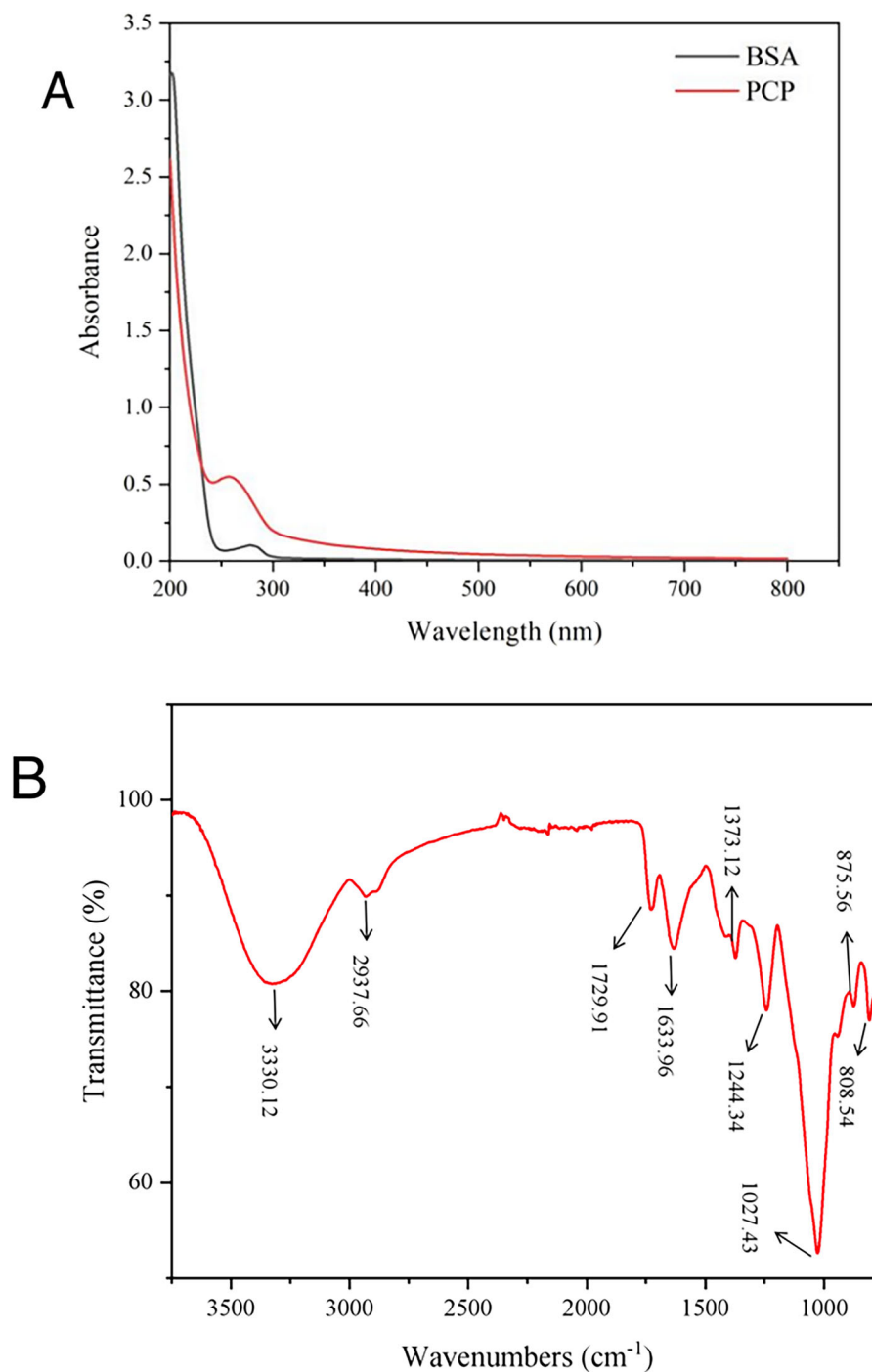


FIGURE 2 | The UV-visible spectrum scanning of PCP (A); the Fourier transform infrared spectroscopy of PCP (B). BSA, bovine serum albumin; PCP, *Polygonatum cyrtoneuma* polysaccharide.

These distinct functional groups drive the sharp reduction in interfacial tension (e.g., from 32.41 to 17.53 mN/m). Furthermore, its moderately high molecular weight (1.27×10^5 Da) and the highly branched, entangled conformation revealed by AFM provide robust steric hindrance and strong inter-chain interactions. These specific structural attributes enable PCP to not only lower interfacial energy but also construct a dense, highly viscoelastic interfacial film, which ultimately serves as the physical foundation for its superior protective effect on encapsulated probiotics.

3.3 | Preparation of HIPEs With Different PCP Concentrations

3.3.1 | Visual Appearance and Microstructural Evolution of PCP-Stabilized HIPEs

HIPEs were fabricated using a gradient of PCP concentrations (1.0–2.5 wt%), with their macroscopic and microscopic features illustrated in Figure 5. At lower concentrations (1.0

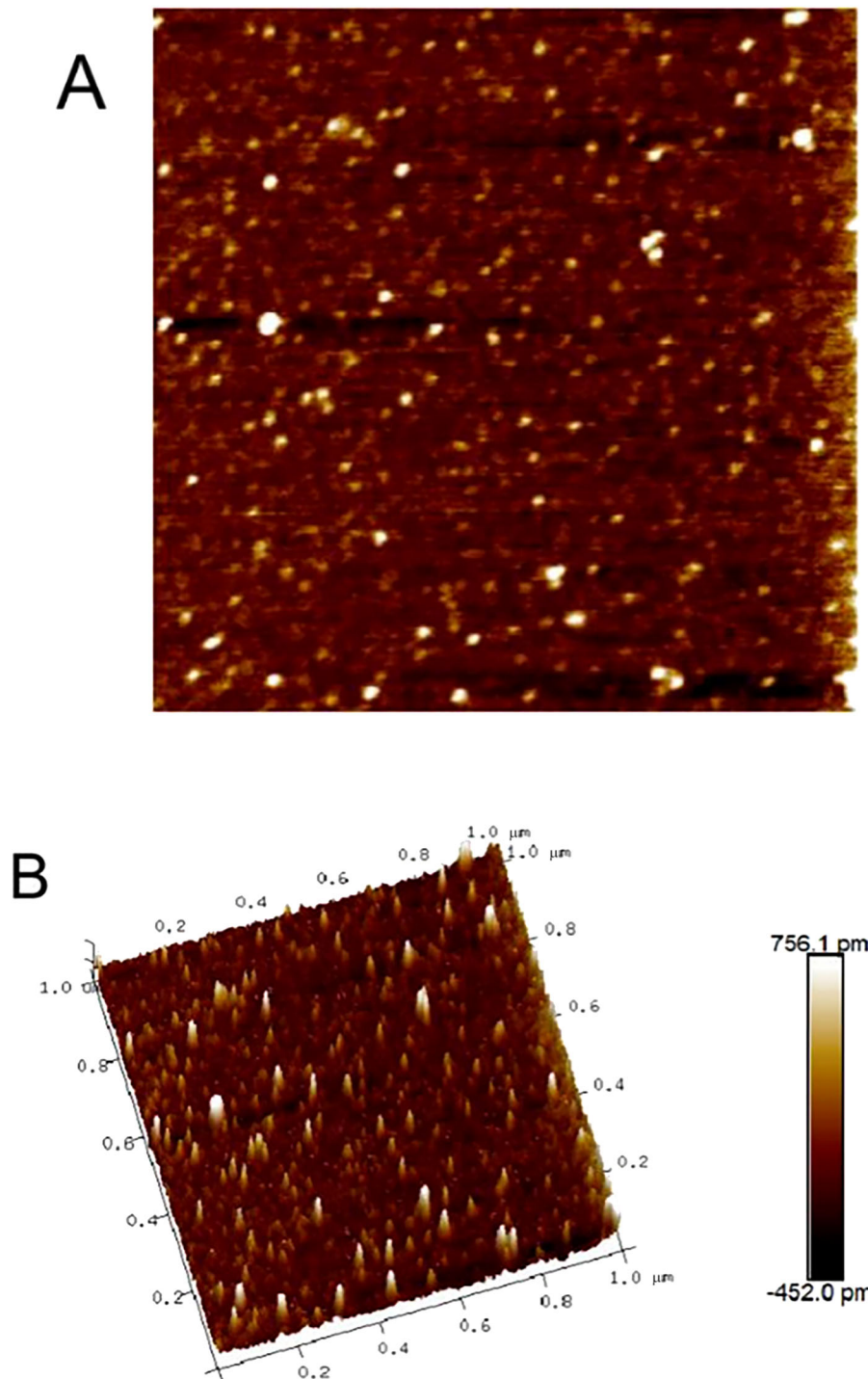


FIGURE 3 | Microstructure morphology (A: 2D, B: 3D) of PCP under AFM.

and 1.5 wt%), the PCP was insufficient to stabilize the high internal phase, resulting in a failure to form a semi-solid gel; these samples exhibited substantial droplet sizes and loose, heterogeneous internal structures. However, HIPEs stabilized with 2.0 and 2.5 wt% PCP transformed into stable, self-standing gels that remained non-flowing upon inversion. These samples displayed a densely packed, multi-faceted polygonal droplet morphology, a hallmark structural feature of well-developed HIPEs (Xu et al. 2025). This transition indicates that a critical PCP concentration is required to provide ade-

quate interfacial coverage and form a robust continuous phase network.

3.3.2 | Average Particle Size and Distribution of HIPEs

Droplet size is a pivotal indicator for evaluating the stability and microstructural refinement of emulsions, where the surface-weighted mean diameter ($D_{3,2}$) reflects the average droplet diameter relative to the total interfacial area. As illustrated in

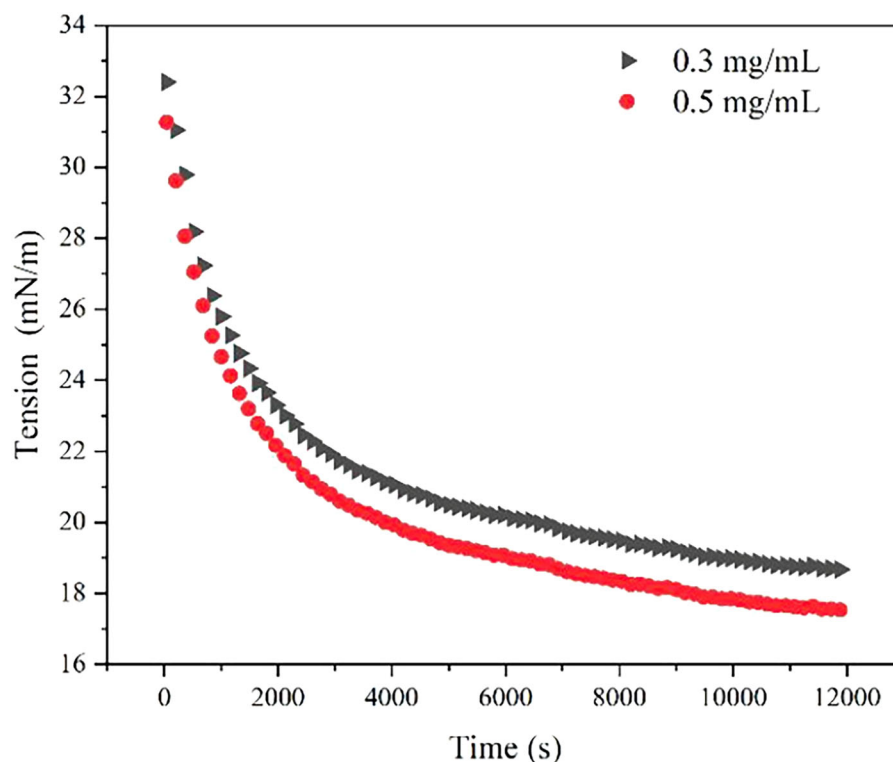


FIGURE 4 | Interfacial tension of different concentrations of PCP at the oil–water interface ($p < 0.05$).

Figure 6A, the HIPEs prepared with 1.0 and 1.5 wt% PCP exhibited significantly larger $D_{3,2}$ values, primarily because these concentrations were insufficient to fully coat the oil droplets, leading to rapid coalescence. In contrast, for the successfully formed HIPEs (2.0–2.5 wt%), the $D_{3,2}$ reached a plateau at 2.0 wt% PCP, where the increased polymer concentration provided adequate interfacial coverage and effectively lowered interfacial tension (Dickinson 2009). Both 2.0 and 2.5 wt% PCP yielded HIPEs with consistently small droplet sizes, with the overlapping size distribution curves (Figure 6B) confirming the high uniformity and reproducibility of these systems. Although the 2.5 wt% sample exhibited a slightly narrower, more symmetrical monomodal distribution, the 2.0 wt% HIPE already demonstrated sufficient stabilization with only a minor shoulder. This minor increase in $D_{3,2}$ at 2.5 wt% could be attributed to the increased aqueous phase viscosity, which might slightly hinder droplet breakup under shear during emulsification (Max et al. 2019). Therefore, 2.0 wt% PCP was selected as the optimal concentration, as it strikes the best balance between interfacial stabilization efficacy and material economy, facilitating the construction of a densely packed and stable microstructure.

3.4 | Protective Effects of HIPEs on *La*

3.4.1 | Macroscopic and Microstructural Characterization of HIPEs

As shown in Figure 7, the macroscopic appearance and microscopic topography of PCP-*La* and GA-*La* HIPEs were character-

ized to evaluate their stability and EE. Macroscopic observations (Figure 7A,D) revealed that both samples formed stable, self-standing gels that remained non-flowing upon inversion; this signifies that the incorporation of *La* did not compromise the robust structural integrity of the HIPE network. Under fluorescence microscopy (Figure 7B,E), both HIPEs exhibited closely packed, multi-faceted polygonal oil droplets, a hallmark of well-developed high internal phase systems (Chen et al. 2020). Importantly, the probiotics (visible as distinct fluorescent puncta) were successfully and uniformly sequestered within the continuous aqueous phase, specifically localized in the interstices between the tightly squeezed oil droplets. This microstructural arrangement confirms that the PCP and GA matrices provide a stable confined environment for the encapsulated bacteria while maintaining a coherent emulsion framework (Sun et al. 2024).

To visualize the spatial distribution and evaluate membrane integrity, dual-fluorescence labeling was conducted using Rhodamine 123 (Rh123) and Nile Red. As seen in Figure 7C,F, *La* within both emulsion systems displayed distinct green fluorescence, confirming intact cellular membranes and metabolic viability. Crucially, the merged images reveal that the viable probiotics are successfully and strictly sequestered within the aqueous interstices between the tightly packed oil droplets (red fluorescence) (Sarode et al. 2021). This confined microstructural arrangement confirms that both PCP and GA matrices provide a robust physical barrier, effectively isolating the encapsulated bacteria from external environmental stressors while maintaining a coherent high internal phase framework.

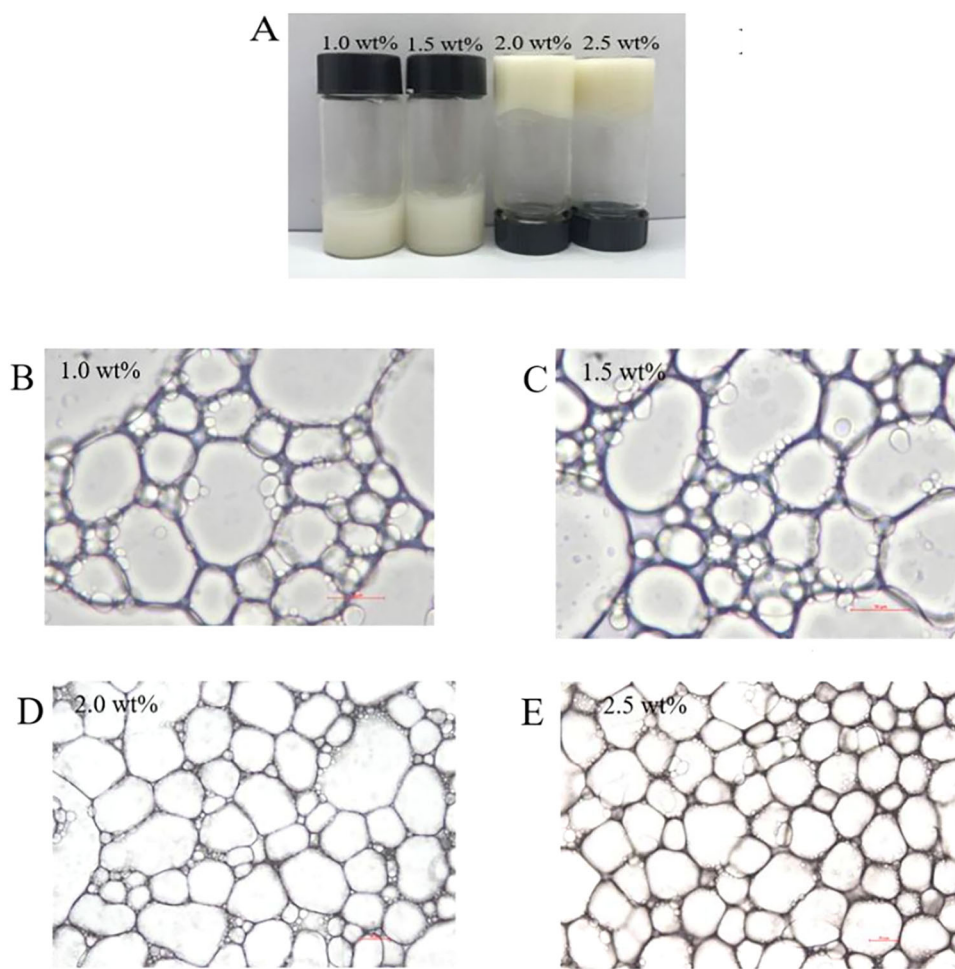


FIGURE 5 | Macro-morphology (A) and microstructure (B–E) of different PCP concentrations (1.0–2.5 wt%) stabilized HIPEs, scale bars: 10 μ m.

3.4.2 | Simulated Gastrointestinal Digestion

Probiotics must maintain a viable concentration to confer health benefits, yet they are easily inactivated by gastric acidity and bile salts (Devarajan et al. 2022). It has been reported that plant-based colloidal delivery systems offer an effective strategy to shield sensitive bioactives, which is a critical requirement for ensuring targeted delivery to the colon (Tan and McClements 2021; Zhong et al. 2025). At the “Fresh” stage (before digestion), the free cell group exhibited a higher initial viable count (9.25 log CFU/mL) compared to the PCP-*La* HIPEs (8.71 log CFU/g) and GA-*La* HIPEs (8.17 log CFU/g). This difference in initial loading is a direct result of the high oil phase volume (80%) in the HIPE formulations, which naturally reduces the density of the aqueous phase containing the probiotics. As shown in Figure 8, the viability of both free and encapsulated *La* declined during simulated digestion (Baranowska-Wójcik et al. 2023). During the gastric phase, free *La* counts dropped precipitously as H^+ influx disrupted intracellular pH homeostasis and enzymatic functions (Martínez et al. 2019). Conversely, PCP-*La* and GA-*La* HIPEs exhibited moderated reductions; the densely packed oil droplets and interfacial films increased the tortuosity of the diffusion path, potentially retarding acid penetration into the aqueous interstices where the bacteria were sequestered.

Upon transition to SIF, PCP-*La* HIPEs maintained significantly higher survival rates than GA-*La* and free *La* groups. Although free cells suffered membrane disruption from bile salts, encapsulated *La* benefited from the protective microenvironment of the HIPE framework. The superior resilience of the PCP-based system compared to traditional matrices (Devarajan et al. 2022; Martínez et al. 2019) is ascribed to its more robust, interwoven polysaccharide network. Consistent with its high viscoelasticity and complex molecular branching, the PCP-based framework provides a superior stable microenvironment compared to simpler linear polysaccharides, such as cellulose or low-viscosity pectin. Although linear polymers often fail to form a sufficiently dense or tortuous network, the branched architecture of PCP facilitates the construction of a more robust physical barrier. This structure effectively isolates cells from the inhibitory effects of bile salts and enzymes, a protective mechanism similar to that observed in highly branched GA or modified starch systems (Sun et al. 2024). Such a dense, entangled matrix limits the diffusion of amphiphilic bile salts, thereby preventing them from interacting with and disrupting the bacterial cell membranes. These findings demonstrate that PCP-based HIPEs function as a high-performance delivery platform, significantly enhancing the gastrointestinal tolerance of *La* and ensuring the delivery of therapeutic doses to the lower GI tract. Although the robust architecture of the PCP-stabilized HIPEs provides excellent isola-

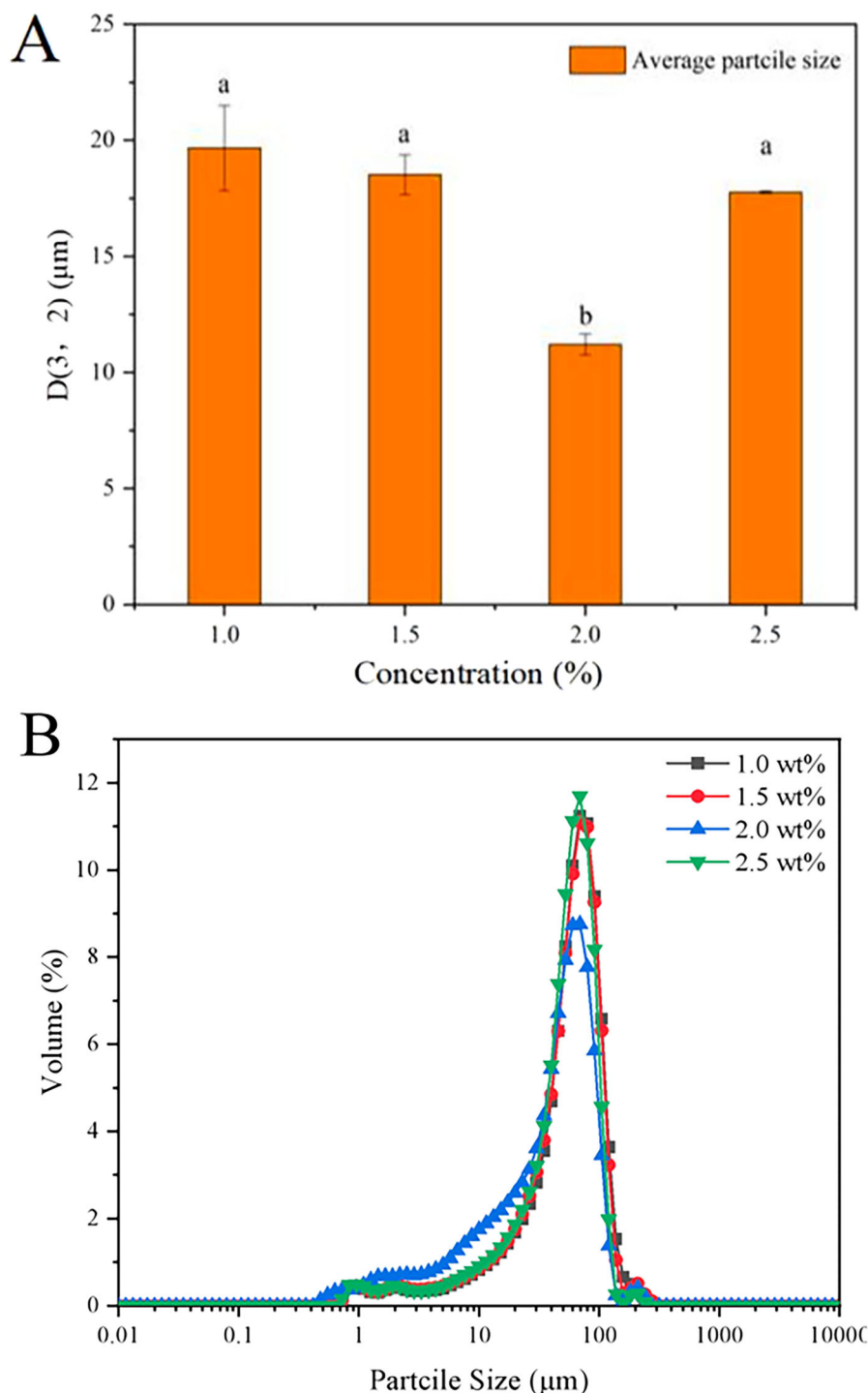


FIGURE 6 | Average particle size ($D(3,2)$) (A), and particle size distribution (B) of PCP HIPEs with different concentrations (1.0–2.5 wt%). Different letters (a, b, c) indicate significant differences ($p < 0.05$).

tion and protection during storage and transit through the hostile upper gastrointestinal tract, the ultimate release of the probiotics is essential for their functional efficacy. It is important to note that this highly stable emulsion matrix is not a permanent barrier but rather a biodegradable delivery vehicle. Upon reaching the lower intestine and colon, the emulsion structure is designed to be dismantled. This physiological breakdown is primarily driven by the enzymatic digestion of the lipid phase by intestinal lipases, coupled with the microbial fermentation of the PCP continuous

phase by the resident colonic microbiota. As the polysaccharide-based protective network is progressively degraded, the entrapped *La* cells are released into the local environment, allowing them to adhere to the gut epithelium, colonize, and effectively exert their associated health benefits.

Building upon these excellent protective and targeted release capabilities, the designed PCP-HIPEs present significant potential for various food industry applications. Specifically, this

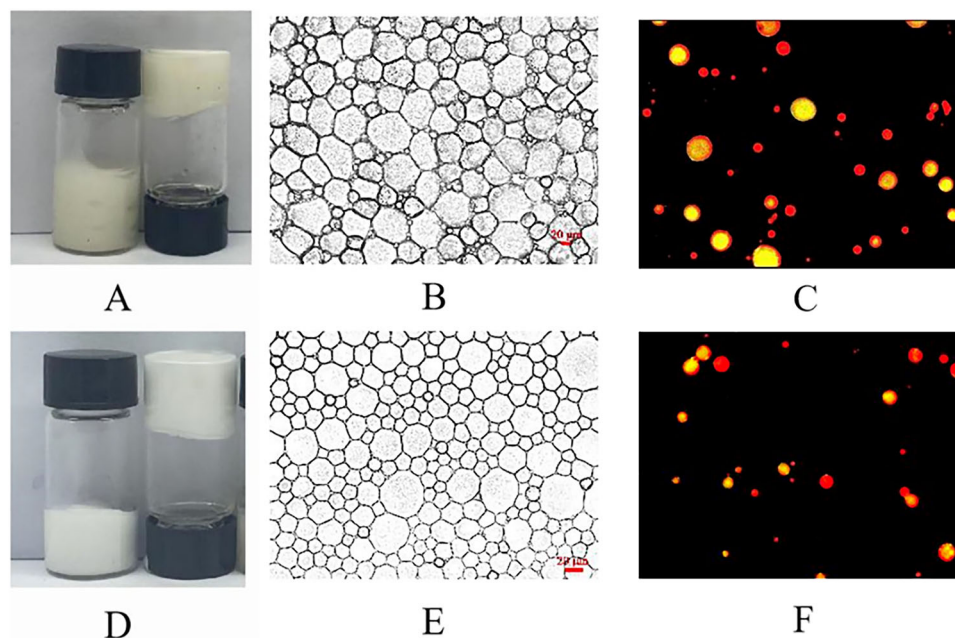


FIGURE 7 | Morphology and probiotic distribution of PCP-*La* and GA-*La* HIPEs. (A and D) Macroscopic gel appearance. (B and E) Microstructure of packed droplets. (C and F) Dual-fluorescence showing viable *La* (green) sequestered within aqueous interstices between oil droplets (red).

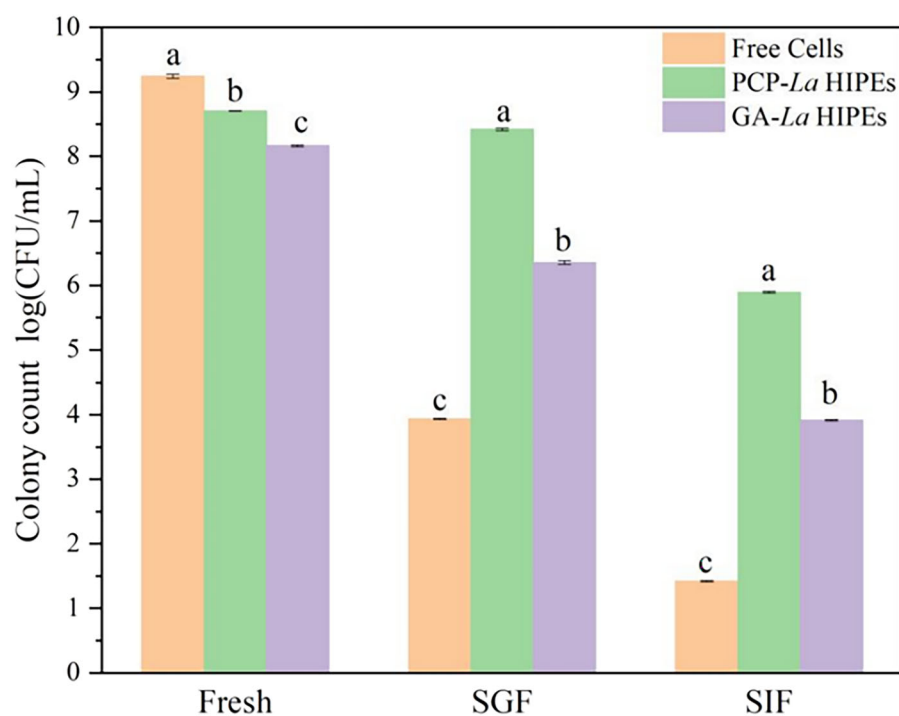


FIGURE 8 | Viability of free *La* (log CFU/mL) and encapsulated probiotics in HIPEs at different GIT digestion stages. Different letters (a–c) indicate significant differences in the groups ($p < 0.05$). GA, gum arabic; HIPE, high internal phase emulsion; *La*, *Lactobacillus acidophilus*; PCP, *Polygonatum cyrtoneuma* polysaccharide; SGF, simulated gastric fluid; SIF, simulated intestinal fluid.

formulation would be highly beneficial for developing functional spreadable foods, such as low-fat mayonnaise alternatives, healthy margarines, and functional plant-based dips. In these products, the HIPE matrix not only serves as a promising fat replacer to reduce caloric and saturated fat intake by mimicking

the desirable rheological properties and mouthfeel of solid fats but also acts as a robust vehicle for probiotic delivery. Furthermore, the protective semi-solid structure makes these emulsions highly suitable for cold-processed desserts, dairy alternatives, and functional bakery fillings, where preserving the viability

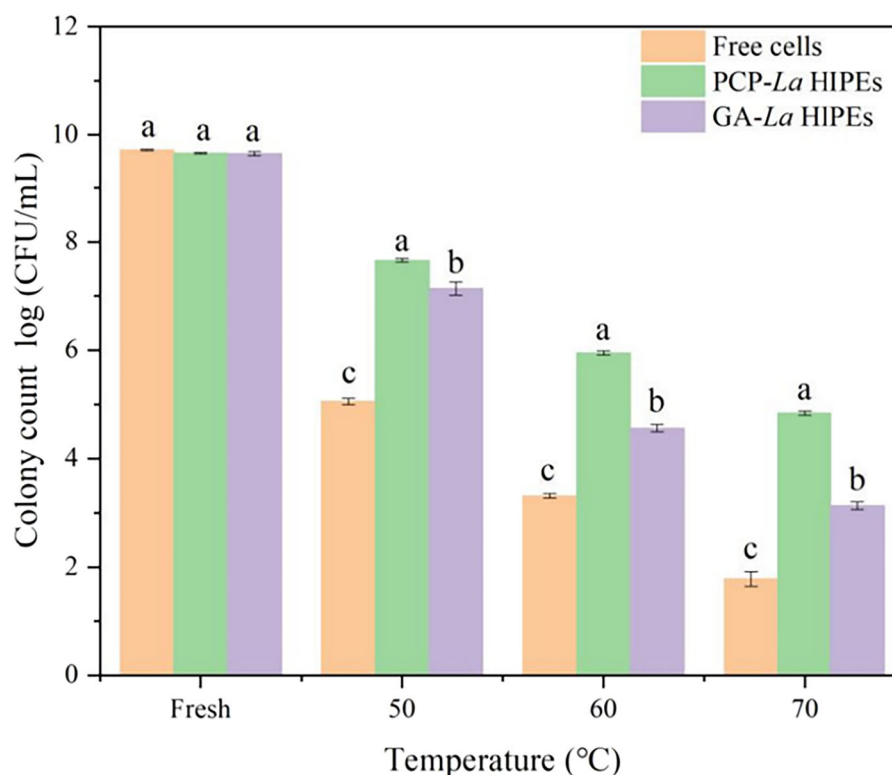


FIGURE 9 | The bacterial viability changes of free and encapsulated *La* (log CFU/mL) in HIEs treated at 50°C, 60°C, and 70°C. Different letters (a–c) indicate significant differences in the groups ($p < 0.05$). GA, gum arabic; *La*, *Lactobacillus acidophilus*; HIE, high internal phase emulsion; PCP, *Polygonatum cyrtoneura* polysaccharide.

of heat-sensitive probiotics like *La* throughout the shelf life is crucial.

3.4.3 | Heat Treatment

The viability of probiotics is frequently compromised by environmental stressors such as elevated temperatures, light, and oxidative conditions (Tripathi and Giri 2014). As illustrated in Figure 9, the thermal stability of free and encapsulated *La* was evaluated at 50°C, 60°C, and 70°C for 30 min. Compared to untreated samples, all groups exhibited a temperature-dependent decline in viable counts. Remarkably, at 50°C, the survival rate of free *La* plummeted by approximately 50%, whereas PCP-*La* and GA-*La* HIEs demonstrated significantly higher resilience, with reductions limited to only 20% and 25%, respectively. This disparity underscores the immediate protective effect of the HIE framework in buffering cells against initial thermal shock.

As the temperature intensified to 60°C and 70°C, the protective efficacy of the two matrices diverged. Although GA-*La* HIEs experienced a substantial loss in activity, PCP-*La* HIEs maintained a markedly more robust survival profile. Compared to other polysaccharide-stabilized systems like modified starch or pectin-based emulsions (Gao et al. 2022; Chen et al. 2021; Sahin et al. 2024), the PCP-based HIE exhibits superior thermal resilience. This is attributed to the dense microstructure and superior interfacial rheology of PCP, which restricts molecular mobility and limits direct heat transfer to the encapsulated

probiotics. These findings suggest that PCP-based HIEs function as an excellent thermal barrier, broadening the application of probiotics in heat-processed functional foods.

3.5 | Comparative Storage Stability of Free and HIE-Encapsulated *La*

3.5.1 | Evolution of HIE Droplet Size During Storage

The storage stability of emulsions is a critical parameter for assessing their long-term performance. Figure 10A,B illustrates that both PCP-*La* HIEs and GA-*La* HIEs maintained a distinct monomodal particle size distribution throughout the 28-day storage period, demonstrating excellent physical stability. Strikingly, the distribution of PCP-*La* HIEs exhibited a slight leftward shift, suggesting the evolution toward a more homogeneous droplet population. This structural integrity can be attributed to the densely packed internal phase of HIEs, where the robust gel networks and elevated continuous phase viscosity effectively suppress droplet coalescence and creaming (Cheng et al. 2021).

Figure 10C reveals that the average droplet size ($D_{3,2}$) for both systems underwent a progressive expansion over time; specifically, PCP-*La* HIEs increased from 12.80 to 24.35 μm , whereas GA-*La* HIEs rose from 23.05 to 28.23 μm . Throughout the storage, GA-*La* HIEs consistently displayed larger droplet sizes than PCP-*La* HIEs. This discrepancy likely stems from the relatively labile interfacial film formed by GA, which is more prone to droplet aggregation. These results collectively indicate

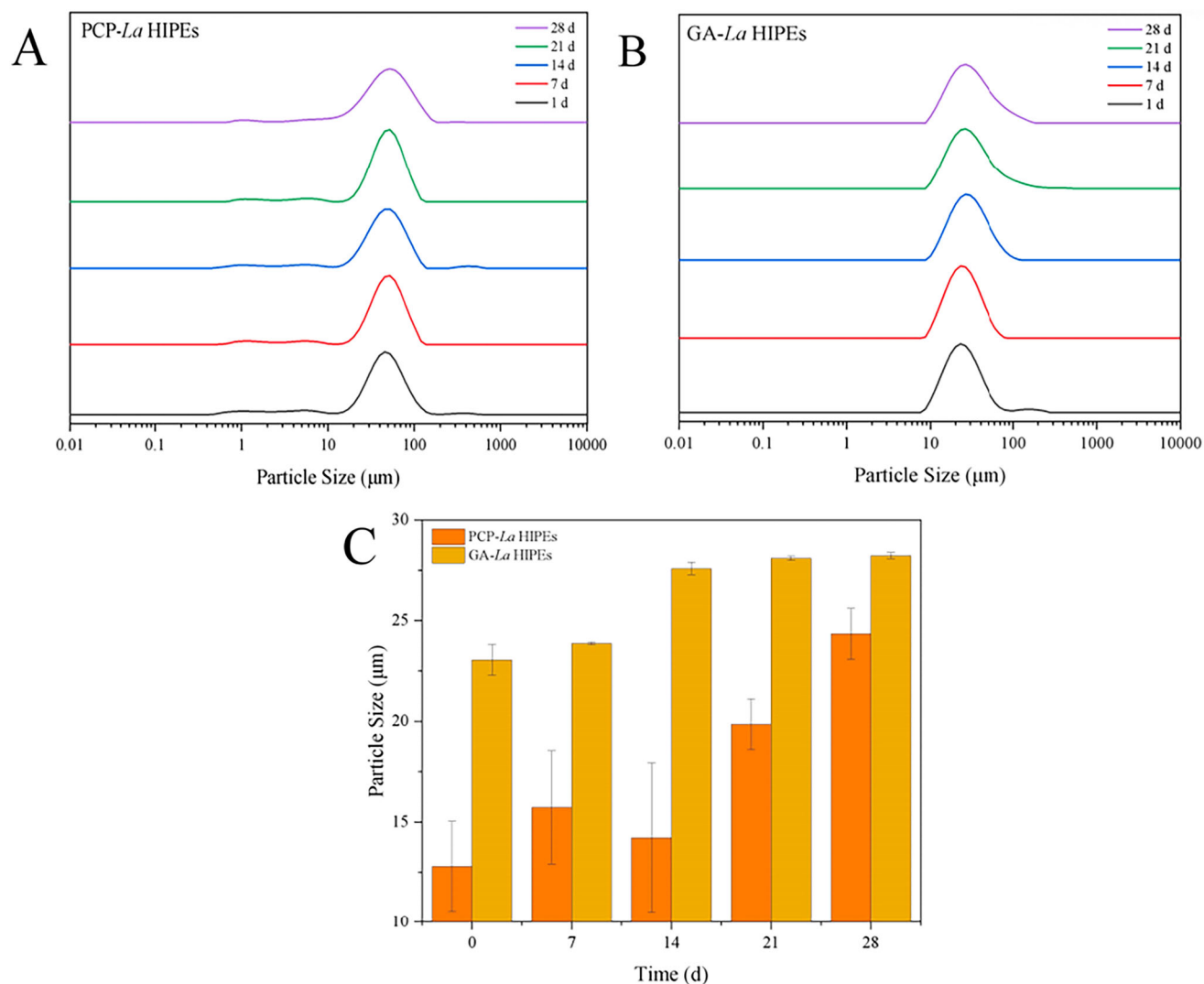


FIGURE 10 | Storage stability of PCP-*La* and GA-*La* HIPEs (4°C, 28 days). (A and B) Monomodal particle size distribution curves. (C) Evolution of mean droplet diameter ($D_{3,2}$), showing better size retention in PCP-stabilized HIPEs. Data are mean $\pm$ SD ($n = 3$). GA, gum arabic; *La*, *Lactobacillus acidophilus*; HIPE, high internal phase emulsion; PCP, *Polygonatum cyrtoneuma* polysaccharide.

that PCP, as an emulsifier, provides superior coverage of oil droplets generated during shearing, thereby more effectively inhibiting droplet coalescence and maintaining emulsion stability compared to GA (Singh et al. 2025). Although PCP-*La* HIPEs exhibited slight statistical fluctuations in during storage, their droplet size remained consistently lower than that of the GA-based system after 28 days (Figure 10C). This variability stems from the high molecular polydispersity of PCP (PDI = 5.531). Crucially, the highly branched and entangled PCP network forms a robust viscoelastic barrier that restricts droplet coalescence more effectively than GA, ensuring superior long-term protection for the encapsulated probiotics.

3.5.2 | Storage Stability

The storage stability of free *La* cells and those encapsulated within PCP-*La* and GA-*La* HIPEs was evaluated over a 28-days period at 4°C. Cell viability was systematically monitored at 7-day intervals (Days 0, 7, 14, 21, and 28). As shown in Figure 11, although

the viable counts in all groups declined during storage, distinct inactivation patterns were observed. Free *La* exhibited the poorest stability, with its population plummeting during the initial 7 days and ultimately reaching the lowest viability by Day 28. This rapid decline underscores the vulnerability of unprotected probiotics to environmental stressors and nutrient depletion (López-Pedrouso et al. 2022). In contrast, both HIPE systems significantly mitigated this loss, maintaining relatively stable counts during the first week. However, as storage progressed beyond 14 days, the viable count in GA-*La* HIPEs began to decrease more rapidly, whereas PCP-*La* HIPEs demonstrated a more gradual and sustained survival trend. By Day 28, PCP-*La* HIPEs retained the highest viable count, significantly outperforming the GA-stabilized system. These results indicate that the PCP-based framework provides a more resilient 3D protective barrier. This dense encapsulation, characterized by high viscoelasticity, potentially restricts cellular metabolic activity and induces a dormant-like state, thereby enhancing long-term resistance to environmental pressure and starvation (Marefati et al. 2021).

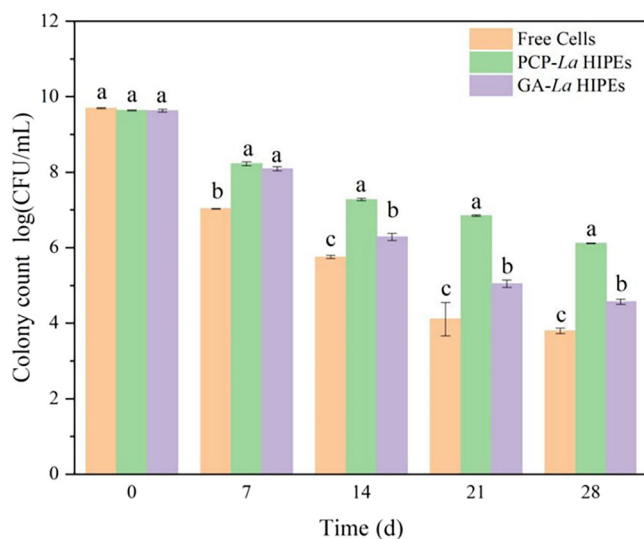


FIGURE 11 | Bacterial viability of free and encapsulated *La* (log CFU/mL) in HIEs at different emulsifiers stored for 28 days at 4°C. Different letters (a–c) indicate significant differences in the groups ($p < 0.05$). GA, gum arabic; *La*, *Lactobacillus acidophilus*; HIE, high internal phase emulsion; PCP, *Polygonatum cyrtonema* polysaccharide.

4 | Conclusions and Prospects

The results of this study demonstrate that PCP, owing to its amphiphilic structure, functions as an effective emulsifier for constructing HIEs and significantly enhances their physical stability. The PCP-stabilized emulsion system effectively improves the gastrointestinal tolerance, thermal stability, and long-term storage survival of the probiotic *La*, demonstrating superior protective performance. Combined with its plant-based origin and the safety profile associated with its traditional use as both food and medicine, PCP shows potential as a natural, clean-label ingredient for functional probiotic delivery systems. It is suitable for applications such as fortified dairy products and shelf-stable probiotic supplements.

On the basis of the current findings, future work could systematically evaluate PCP against a broader range of emulsifiers, including diverse plant polysaccharides, modified starches, and synthetic options, to better define its practical advantages and economic feasibility within complex food formulations. Future studies could further elucidate the relationship between the PCP interfacial architecture and the stress resistance of encapsulated probiotics. Additionally, evaluating the performance of PCP-stabilized HIEs in diverse food matrices and under industrial processing conditions would be beneficial for assessing their practical application potential in the functional food industry. Despite the demonstrated protective efficacy of PCP-stabilized HIEs under controlled laboratory conditions, several challenges must be addressed to fully establish their potential for commercial functional food applications. The industrial scale-up is currently constrained by the extraction yield and associated costs of natural PCP, which can fluctuate depending on the geographical origin, harvest season, and traditional extraction methods of *P. cyrtonema*. Furthermore, translating the high-shear homogenization process to an industrial scale requires significant energy consumption and specialized continuous emulsification

equipment to maintain the structural integrity and uniformity of the HIEs. Therefore, future studies should focus on optimizing high-efficiency and green extraction technologies (e.g., ultrasound- or enzyme-assisted extraction) to enhance PCP yield and reduce raw material costs. Exploring continuous processing methods and evaluating the cost-effectiveness of these HIE-based probiotic delivery systems will be crucial steps toward their successful integration into the food industry.

Author Contributions

Zhao Wang: investigation, writing – original draft. **Zhifa Zhang:** investigation. **Chunyan Peng:** data curation. **Yihui Dai:** visualization. **Yi Huang:** conceptualization. **Xiaoqiang Chen:** conceptualization, methodology, writing – review and editing, supervision.

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Conflicts of Interest

The authors declare no conflicts of interest.

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